Bellevue, WA
Phone: 412-522-3541
Portfolio: https://sohhyy.github.io/game/
Email: bokangw@alumni.cmu.edu

Bokang Wang

EDUCATION

Carnegie Mellon University, Entertainment Technology Center (ETC) Pittsburgh, PA
Master of Entertainment Technology *2021 - 2023*

The Hong Kong Polytechnic University Hong Kong
Bachelor of Science in Computing *2016 - 2020*

SKILLS

Languages: C#, C++, C, Java, JavaScript, HTML, CSS, Python
Tools: Unity, Unreal, Visual Studio, Git, Source Tree, Perforce, 8Th Wall
Game Development: VR/AR, Gameplay System, Performance Optimization, Rapid Prototyping

WORK EXPERIENCE

Rec Room Inc Seattle, WA
Software Engineer (Unity/C#) *Jan.2023. – Aug.2025*

- Developed over 30 user-generated content (**UGC**) chips and tools that enable core **gameplay mechanics** in UGC room, including core systems such as Interaction Volume, Custom Footstep Sound, Dialogue UI, Inventory UI Animation, and multiple 3D math chips; collaborated closely with designers to translate gameplay concepts into robust features.
- Redesigned of the **MakerPen mobile UI,** Rec Room's in-game creation tool, using **MVVM architecture**, making it more intuitive, reducing input complexity, and boosting productivity for creators on mobile devices.
- **Optimized memory usage** across three first-party games *(Make it to Midnight*, *My Little Monster*, *Run the Block*), **reducing memory usage by ~10% and lowering crash rate by ~5%** on low-end mobile devices.
- Built a **custom Unity tool** to analyze and visualize memory usage of in-scene UGC chips and tools, helping designers and engineers to quickly identify and resolve memory bottlenecks.
- Implemented **client-side logic** for custom dorms, allowing users to rename and update their dorm rooms. Worked closely with backend engineers **exchanging JSON data** with the backend to synchronize player updates.
- Implemented Magnetic Snapping, a new object alignment system that enables players to snap objects together via magnetic points, making in-game building more intuitive. Built all the underlying **3D math logic** for this feature.

Orta Interactive Studio Pittsburgh, PA
VR Software Engineer Intern (Unity/C#) *Jun. – Aug.2022*

- Developed *Romeo and Juliet - VR*, an interactive game to educate opera performers on acting and line-reading.
- Recorded animation using the **Motion Capture** system and provided technical support to artists for processing data
- Implemented all core gameplay systems including player controller, UI, animation, and lighting in Unity.

PROJECTS

Theater ARchive, La MaMa Experimental Theatre Club New York, NY
Web AR Developer *Sep. – Dec. 2022*

- Developed an immersive **Web AR archive experience** on Niantic's 8th Wall platform as the solo programmer in a 5-member team to enhance the digital presentation of La MaMa's archival collections. And it was showcased by La MaMa Theatre as part of their digital archive initiative.
- Implemented the entire front-end with **HTML/CSS**, ensuring cross-device compatibility and smooth performance.
- Designed and developed five interactive features, including a face filter for virtual masks, body tracking to embody stage puppets, and image recognition that made posters, props, and costumes interactive.

3D Data Visualization of Machine Learning, Google & ETC Pittsburgh, PA
Producer, Programmer (Unity/C#) *Jan. – May. 2022*

- **Led a team of 7**, served as the point of contact with advising professor and Google scientists, fostered productive team communication, managed the development process, and maintained development documentation and website
- Visualized 4 types of data in 3D environments through 5 interactive experiences with 3 ML algorithms (K-means, Q-learning, Fast Fourier transform) and 3 platforms (PC, VR. Ultrahaptic)